

ECON 524 – PROBLEM SET 1

1. Recall that one way to measure the performance of prediction is

$$\sqrt{E_X[(X^\top \hat{\beta}_n - X^\top \beta_0)^2]} \quad (1)$$

where $\hat{\beta}_n$ is some estimator $\hat{\beta}_n$ of β_0 (e.g., OLS or LASSO).

- (a) Suppose that $X = x_j$ with probability p_j for $j = 1, \dots, 5$ such that $p_1 + \dots + p_5 = 1$. Compute (1) in this case.
 - (b) When $\hat{\beta}_n$ is the LASSO estimator, the upper bound of (1) we learned in class is $\sqrt{s \log(n \vee p)/n}$, which is different from the bound $\sqrt{p/n}$ when $\hat{\beta}_n$ is the OLS estimator. Explain the differences.
 - (c) Suppose that $s = n^b$ and $p = e^{n^c}$ for some $b, c > 0$ (we ignore the fact that s and p can only be integers). Under what condition(s) do we have (1) tending to zero for the LASSO case?
2. We have studied the ridge regression, LASSO, and the elastic net as three ways of handling high dimensional linear regression.
- (a) How do they compare to OLS in terms of the bias-variance trade-off?
 - (b) All three methods perform shrinkage. How do they compare to one another when it comes to shrinkage?
 - (c) Do they all perform variable selection? Explain.
3. There are two versions of LASSO discussed in class.
- (a) What are the motivations of the robust LASSO developed by Belloni, Chen, Chernozhukov, and Hansen (2012)?
 - (b) Explain how and why post-LASSO improves (robust) LASSO.
4. Logistic regression and support vector classifier are two important classification methods in machine learning.
- (a) Show that, for any pair (α, β) ,

$$\begin{aligned} \log\left(\left[\frac{\exp(\alpha + X_i^\top \beta)}{1 + \exp(\alpha + X_i^\top \beta)}\right]^{1\{Y_i=1\}} \left[1 - \frac{\exp(\alpha + X_i^\top \beta)}{1 + \exp(\alpha + X_i^\top \beta)}\right]^{1\{Y_i=0\}}\right) \\ = Y_i\{\alpha + X_i^\top \beta\} - \log(1 + \exp\{\alpha + X_i^\top \beta\}) . \end{aligned} \quad (2)$$

- (b) Suppose we collect data for a group of students in a statistics class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficients, $\hat{\alpha}_{0,n} = -6$,

$\hat{\beta}_{1,n} = 0.05$, and $\hat{\beta}_{2,n} = 1$. Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class. How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

- (c) What are the main motivations of the support vector classifier as a soft margin classifier?